Que:- 1 Write a program that demonstrates the order of execution of operators in expressions

```
#include <stdio.h>
int main() {
    int a = 5, b = 10, c = 2, result1, result2;

    result1 = a + b * c;
    result2 = (a + b) * c;

    printf("a + b * c          = %d\n", result1);
    printf("(a + b) * c        = %d\n", result2);

    int x = 5, y = 10, z = 15;
    int logicResult = x < y && y < z;

    printf("x < y && y < z     = %d\n", logicResult);

    return 0;
}
```

OutPut

```
a + b * c          = 25
(a + b) * c        = 30
x < y && y < z     = 1
```

Que; 2 Write a program to check whether a given number is even or odd

```
#include <stdio.h>

int main() {
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);

    if (number % 2 == 0) {
        printf("%d is Even\n", number);
    } else {
        printf("%d is Odd\n", number);
    }

    return 0;
}
```

OutPut

```
Enter a number: 8
8 is Even
```

Que.3:- Write a program to determine whether a given number is positive, negative, or zero.

```
#include <stdio.h>

int main() {
    int number;

    printf("Enter a number: ");
    scanf("%d", &number);

    if (number > 0) {
        printf("%d is Positive\n", number);
    } else if (number < 0) {
        printf("%d is Negative\n", number);
    } else {
        printf("The number is Zero\n");
    }

    return 0;
}
```

OutPut

```
Enter a number: -5
-5 is Negative
```

Que.5:- Write a program to generate the multiplication table of any number using while loop, do while loop and for loop.

```
#include <stdio.h>

int main() {
    int n, i;

    printf("Enter a number: ");
    scanf("%d", &n);

    printf("\nMultiplication Table using while loop:\n");
    i = 1;
    while (i <= 10) {
        printf("%d x %d = %d\n", n, i, n * i);
        i++;
    }

    printf("\nMultiplication Table using do-while loop:\n");
    i = 1;
    do {
        printf("%d x %d = %d\n", n, i, n * i);
        i++;
    } while (i <= 10);

    printf("\nMultiplication Table using for loop:\n");
    for (i = 1; i <= 10; i++) {
        printf("%d x %d = %d\n", n, i, n * i);
    }

    return 0;
}
```

OutPut

Enter a number: 5

Multiplication Table using while loop:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Multiplication Table using do-while loop:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50

Multiplication Table using for loop:
5 x 1 = 5
5 x 2 = 10
5 x 3 = 15
5 x 4 = 20
5 x 5 = 25
5 x 6 = 30
5 x 7 = 35
5 x 8 = 40
5 x 9 = 45
5 x 10 = 50